Systematic review: factors associated with risk for and possible prevention of cognitive decline in later life.

BACKGROUND: Many biological, behavioral, social, and environmental factors may contribute to the delay or prevention of cognitive decline. PURPOSE: To summarize evidence about putative risk and protective factors for cognitive decline in older adults and the effects of interventions for preserving cognition. DATA SOURCES: English-language publications in MEDLINE, HuGEpedia, AlzGene, and the Cochrane Database of Systematic Reviews from 1984 through 27 October 2009. STUDY SELECTION: Observational studies with 300 or more participants and randomized, controlled trials (RCTs) with 50 or more adult participants who were 50 years or older, drawn from general populations, and followed for at least 1 year were included. Relevant, good-quality systematic reviews were also eligible. DATA EXTRACTION: Information on study design, outcomes, and quality were extracted by one researcher and verified by another. An overall rating of the quality of evidence was assigned by using the GRADE (Grading of Recommendations Assessment, Development, and Evaluation) criteria. DATA SYNTHESIS: 127 observational studies, 22 RCTs, and 16 systematic reviews were reviewed in the areas of nutritional factors; medical factors and medications; social, economic, or behavioral factors; toxic environmental exposures; and genetics. Few of the factors had sufficient evidence to support an association with cognitive decline. On the basis of observational studies, evidence that supported the benefits of selected nutritional factors or cognitive, physical, or other leisure activities was limited. Current tobacco use, the apolipoprotein E epsilon4 genotype, and certain medical conditions were associated with increased risk. One RCT found a small, sustained benefit from cognitive training (high quality of evidence) and a small RCT reported that physical exercise helps to maintain cognitive function. LIMITATIONS: The categorization and definition of exposures were heterogeneous. Few studies were designed a priori to assess associations between specific exposures and cognitive decline. The review included only English-language studies, prioritized categorical outcomes, and excluded small studies. CONCLUSION: Few potentially beneficial factors were identified from the evidence on risk or protective factors associated with cognitive decline, but the overall quality of the evidence was low. PRIMARY FUNDING SOURCE: Agency for Healthcare Research and Quality and the National Institute on Aging, through the Office of Medical Applications of Research, National Institutes of Health.